DATA CLEANING & ANALYSIS LOG

Retail Store Sales — Portfolio Project

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Dataset Source: Kaggle — Retail Store Sales

Date Cleaned: July 2026

Tool Used: Microsoft Excel

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1. Dataset Overview

Detail	Value
Total Columns	11 (A–K)
Total Data Rows	12,575 (12,576 including header row)
Confirmed Via	Ctrl+End shortcut in Excel

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2. Missing Values — Inspection (via COUNTBLANK)

Column	Blanks Found	Action Taken	Reason
Transaction ID	0	None	Clean
Customer ID	0	None	Clean
Category	0	None	Clean
Item	95	Filled with "Unknown"	No reliable way to infer item name — Unknown preserves the row without guessing
Price Per Unit	47	Recalculated: Total Spent / Quantity	Mathematically recoverable from the other two linked columns
Quantity	48	Recalculated: Total Spent / Price Per Unit	Mathematically recoverable from the other two linked columns
Total Spent	48	Recalculated: Price Per Unit x Quantity	Mathematically recoverable from the other two linked columns
Payment Method	0	None	Clean
Location	0	None	Clean
Discount Applied	4,199	Filled with FALSE	Blank assumed to mean no discount applied — FALSE is the correct logical value

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3. Data Type Check

Check	Method Used	Result
Numeric columns stored as text	ISNUMBER formula + Text to Columns	No issues — all numeric fields confirmed as true numbers

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4. Casing & Consistency Check (via Pivot Table)

Column	Method	Unique Values Found	Result
Category	Pivot Table	8 (Beverages, Butchers, Computers and electric accessories, Electric household essentials, Food, Furniture, Milk Products, Patisserie)	No inconsistencies
Payment Method	Pivot Table	—	No inconsistencies

Column	Method	Unique Values Found	Result
Location	Pivot Table	—	No inconsistencies

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5. Calculated Field Integrity

Price Per Unit, Quantity, and Total Spent are mathematically linked:

Price Per Unit × Quantity = Total Spent

Missing values were cross-recalculated using IF formulas in helper columns, then pasted back as plain values using Paste Special → Values:

Missing Total Spent:

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=IF(G2="", E2*F2, G2)
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Missing Price Per Unit:

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=IF(E2="", G2/F2, E2)
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Missing Quantity:

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=IF(F2="", G2/E2, F2)
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6. Duplicate Check

Check	Method	Result
Duplicate transactions	Data → Remove Duplicates → Transaction ID only	0 duplicates found — all 12,575 rows are unique

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7. Final Dataset Summary

Detail	Before Cleaning	After Cleaning
Total Rows	12,575	12,575 (no rows removed)
Blank Cells	4,485 across all columns	0
Text-Formatted Numbers	0 found	—
Casing Inconsistencies	0 found	—
Duplicate Transactions	0 found	—

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8. Cleaning Decisions Summary

- No rows were deleted — all missing values were either recalculated or labeled, preserving the full dataset of 12,575 rows.
- Discount Applied blanks (4,199) were treated as FALSE rather than Unknown — blank logically means no discount was applied.
- Item blanks (95) were labeled Unknown since item name cannot be inferred from any other column.
- Price Per Unit, Quantity, and Total Spent blanks were recovered using their mathematical relationship, not estimates or averages.
- All three categorical columns (Category, Payment Method, Location) were clean — no casing or spacing fixes required.

9. Exploratory Data Analysis (EDA)

5 business questions were analysed using Pivot Tables in Microsoft Excel. All findings are based on the fully cleaned dataset of 12,575 rows.

Q1: Which category generates the most revenue?

Category	Total Revenue
Butchers	208,118 (highest)
Electric household essentials	203,813.50
Beverages	197,047.50
Furniture	195,310
Food	194,812
Computers and electric accessories	190,692.50
Patisserie	182,165.50
Milk Products	180,112 (lowest)

Insight: Butchers is the top revenue-generating category at 208,118, roughly 15% higher than the lowest category (Milk Products at 180,112). The revenue gap between all categories is relatively small, indicating a well-balanced product mix with no single dominant category.

Q2: Which sales channel performs better?

Channel	Total Revenue
Online	791,401 (51%)
In-Store	760,670 (49%)
Grand Total	1,552,071

Insight: Online sales marginally outperform in-store sales by approximately 4%. The near-even split suggests the business has a strong physical retail presence while also performing well digitally — a healthy dual-channel balance.

Q3: Which payment method is most popular?

Payment Method	Transaction Count	Share
Cash	4,310	34.3%
Digital Wallet	4,144	33.0%
Credit Card	4,121	32.8%
Grand Total	12,575	100%

Insight: Cash is the most popular payment method but only by a very small margin — the three methods are almost evenly distributed across all transactions. The gap between first (Cash) and last (Credit Card) is only 189 transactions out of 12,575, suggesting customers have no strong payment preference.

Q4: Do discounts increase spending?

Discount Applied	Total Revenue	Transactions	Avg. Spend Per Transaction
FALSE (No discount)	1,027,627.50	8,356	~122.97
TRUE (Discount applied)	524,443.50	4,219	~124.30

Insight: Although discounted transactions are fewer (4,219 vs 8,356), they generate a slightly higher average spend per transaction (124.30 vs 122.97). This suggests discounts may encourage customers to spend marginally more per visit. However the difference is small and would benefit from further statistical testing.

Q5: Which category sells the most units?

Category	Units Sold
Furniture	8,462 (highest)
Food	8,387
Beverages	8,358
Milk Products	8,339
Electric household essentials	8,309
Computers and electric accessories	8,272
Butchers	8,206
Patisserie	7,943 (lowest)
Grand Total	66,276

Insight: Furniture sells the most units (8,462) but ranks 4th in revenue — while Butchers generates the most revenue but ranks 7th in units sold. This reveals that Butcher items have a higher price per unit, while Furniture items are purchased more frequently at lower prices. This is a key distinction between volume-driven and value-driven categories.

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10. Key Insights Summary

#	Question	Finding
1	Top revenue category	Butchers — 208,118
2	Best sales channel	Online — 791,401 (51% of total revenue)
3	Most popular payment	Cash — 4,310 transactions (34.3%)
4	Discount impact	Discounted transactions average slightly more per sale (124.30 vs 122.97)
5	Most units sold	Furniture — 8,462 units, but lower revenue per unit than Butchers